

YOUNGYOON CHOI

+82(10) 3487-3558 ◇ youngyoon911@snu.ac.kr ◇ youngyoon911.github.io

RESEARCH INTERESTS

Learning the world from visual experience

- **Generative Modeling** – Diffusion & flow models, controllable generation, representation learning
- **Visual World Models** – Toward long-horizon, world-consistent, and interactive generation

PAPERS

Diverse Motion Customization via Control-based Dynamic Optimization

Youngyoon Choi*, Kihyun Kim*, Jeongwoo Shin, Joonseok Lee

Under review, 2026

Keywords: Motion Customization, Stochastic Optimal Control, Controllable video generation

Trajectory-guided Referred Motion Segmentation

Yunjeong Choi, Hyunjin Lee, **Youngyoon Choi**, Eunyi Lyoo, Yoori Oh, Joonseok Lee

Under review, 2026

Keywords: Referring Video Object Segmentation, Point Trajectory, Motion

Improving 3D MRI Generation via Latent Space Decorrelation

Wonyoung Jang*, Junho Lee*, **Youngyoon Choi**, Seunggeun Lee, Joonseok Lee

Medical Image Understanding and Analysis (MIUA) 2026, **Oral**

Keywords: Brain Imaging, Flow matching, Representation Learning

GaussianVideo: Efficient Video Representation and Compression by Gaussian Splatting

Inseo Lee, **Youngyoon Choi**, Joonseok Lee

CVPR Workshop on Perception Beyond the Visible Spectrum 2025

Keywords: 3D Gaussian Splatting, Video Compression, Neural Rendering

EDUCATION

M.S. in Graduate School of Data Science

Mar 2024 – Present

Seoul National University, South Korea

Advisor: Joonseok Lee · Expected Feb 2027

B.A. in Psychology (Interdisciplinary Program in Brain, Mind, and Behavior)

Mar 2019 – Feb 2024

Seoul National University, South Korea

CGPA: 3.96 / 4.3 · Degree Honors: Summa Cum Laude

RESEARCH EXPERIENCE

Graduate Researcher, Visual Information Processing (VIP) Lab

Feb 2024 – Present

Advisor: Joonseok Lee

- Conducting research on controllable video generation grounded in theory-driven approaches.
- Recent work applies stochastic optimal control to address content leakage in motion customization.

Undergraduate Researcher, Computational Clinical Science (CCS) Lab

Sep 2023 – Dec 2023

Advisor: Wooyoung Ahn

- Conducted research on multimodal decision-making prediction, applying probabilistic modeling to integrate heterogeneous data.
- Received 3rd place at a research symposium.

Undergraduate Researcher, Human Factors Psychology (HFP) Lab

Jan 2023 – May 2023

Advisor: Sowon Hahn

- Assisted in participant recruitment and experimental procedures for a study examining how feedback modality in human-robot interaction affects exercise behavior change.

Undergraduate Researcher, Connectome Lab

Jan 2022 – Dec 2022

Advisor: Jiwook Cha

- Conducted research on youth suicide, applying machine learning to identify risk factors.
- Selected for the Undergraduate Research Funding Program.

SELECTED PROJECTS

LexiMate: An End-to-End Inclusive System for Dyslexia Support

Developer / Data Engineer / Domain Expert

- Built a product-level end-to-end system for dyslexia screening and intervention using real-time multimodal data.
- Designed and developed the data collection pipeline, enabling structured acquisition of heterogeneous multimodal inputs.

Efficient Motion Generation with Mean Flows

Team Researcher

- Developed a one-step text-conditioned 3D human motion generation framework using mean flow modeling, achieving real-time synthesis with reduced inference cost over diffusion-based methods.
- Collaborated on designing the continuous-time vector field model in latent motion space.

Advancing Image Reconstruction: Evaluating BOLD and Beta Signals in fMRI-Based Brain Decoding

Team Researcher

- Explored diffusion-based generative approaches for fMRI-to-image reconstruction, evaluating output quality via CLIP, SSIM, and semantic similarity.
- Designed and ran the multi-metric evaluation pipeline to quantitatively compare BOLD signal- and beta value-based reconstructions.

Honorable Military Family Benefits Platform

Backend Developer

- Built a backend system enabling personalized benefit matching for eligible users using PostgreSQL.
- Contributed as the primary backend developer, handling database design and query optimization.

TEACHING EXPERIENCE

Basic Computing: First Adventures in Computing

Feb 2024 – June 2024

Led weekly hands-on Python programming classes for undergraduate students

Computational Thinking: Thinking with Computers

Dec 2025

Delivered a special lecture on data analysis and visualization

Machine Learning and Deep Learning I

Sep 2024 – Nov 2024

SCHOLARSHIPS & AWARDS

4th stage BK21 Scholarship

Sep 2024 – Dec 2025

3rd Place, Brain-Mind-Behavior Interdisciplinary Research Presentation

Dec 2023

Undergraduate Research Funding Program

Jul 2022 – Dec 2022

Dean's List, Academic Scholarship

Dec 2020

SKILLS

Languages Korean (Native), English (Native), Spanish (Functional)

Programming Python, R, SQL, C/C++